# A Numerically Resolved Free-Boundary MOTS Saddle-Node in Dynamical Braneworld Collapse

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**Abstract**

We numerically resolve the critical structure of a brane-terminating marginally outer trapped-surface (MOTS) pair in the $SO(3)$-symmetric sector of five-dimensional Einstein–anti-de Sitter gravity on a Goldberger–Wise-stabilized compact interval. An independently constructed parent sequence contains no accepted upper-brane-capped MOTS at $t/\ell = 0.001$ and exactly two at $t/\ell = 0.0015$ on three successive spacetime grids. Pseudo-arclength continuation of the coupled surface–time problem locates a common critical time $t_*/\ell \simeq 1.17254 \times 10^{-3}$ in the chosen foliation. The principal stability eigenvalue crosses zero with a real, sign-definite mode, while adjoint projections give nonzero near-critical transversality and quadratic coefficients under a fully specified discrete normalization. The invariant one-sided proper-area separation follows the expected square-root law. An independent G10 half-timestep evolution repeats the parent and continuation, yields an overlapping critical-time bracket, and gives a free exponent $p \simeq 0.498$. These results numerically resolve the established saddle-node mechanism in a dynamical free-boundary MOTS problem rather than inferring it only from pair anatomy. The claim is finite-resolution and quasi-local: it is not a continuum theorem, event-horizon reconstruction, or phase diagram.

## 1 Introduction

Black objects in compact warped geometries probe localization, stabilization, and boundary dynamics simultaneously [1, 2, 3, 4]. Static braneworld calculations have constructed localized solutions from small to large horizon radii [5, 6]; time-symmetric data have produced cigar-shaped apparent horizons, and full Randall–Sundrum II collapse has produced brane-intersecting horizons of finite bulk extent [7, 8]. These results establish that a brane need not confine the relevant horizon geometry to the induced four-dimensional spacetime. They do not, however, determine which free-boundary marginal surfaces are dynamically selected in a stabilized compact two-brane evolution.

Marginally outer trapped surfaces are quasi-local and slicing-dependent, but their defining equation is geometric on each slice and their stability operator organizes local continuation of branches [9, 10, 11, 12]. Multiple MOTSs, branch creation, and alternating stability are established features of four-dimensional black-hole evolutions [13, 14, 15, 16]. Higher-dimensional MOTSs and free-boundary or capillary MOTSs have independent mathematical treatments, including higher-dimensional stability, curvature, and rigidity results [17, 18, 19, 20, 21]. In that literature, orthogonal contact with the support hypersurface is the free-boundary condition; the brane endpoint condition used below is precisely this orthogonality condition. Recent bifurcation analysis identifies generic pair creation as a saddle-node mechanism and distinguishes it from symmetry-supported pitchfork or transcritical cases [22, 23]. Accordingly, this paper does not claim the fold mechanism as new. Its question is whether the mechanism can be resolved—rather than recognized only from its aftermath—for brane-terminating free-boundary MOTSs in a compact, explicitly stabilized braneworld.

The new result is a normal-form structure reproduced across three spatial grids and in an independent half-timestep evolution in a substantially different operator problem. The spacetime is generated by a nonlinear five-dimensional initial-boundary-value evolution rather than prescribed as an exact metric family; the MOTSs have a physical boundary; the stability operator is non-self-adjoint; and its domain contains the Robin condition induced by the brane geometry. Existing closed-MOTS bifurcation examples do not contain these features, while existing free-boundary stability and rigidity results do not guarantee dynamical pair creation. Thus we test the reach of the general MOTS bifurcation framework in a setting where its realization was unresolved, without proposing a new universal bifurcation class.

We answer that question by combining direct three-grid formation with pseudo-arclength continuation in surface profile and time. The two branches coalesce at a grid-consistent critical surface, their principal eigenvalue is bracketed through zero, an isolated near-critical principal mode is separated from the remainder of the reduced-sector spectrum, the adjoint transversality and quadratic coefficients are nonzero, and the invariant area separation has the expected square-root exponent. An independent G10 half-timestep evolution repeats the complete parent, continuation, and critical analysis, while operator-resolution, boundary, residual, and replay controls test the principal numerical alternatives. The central claim is a finite-resolution numerical resolution, in an $SO(3)$-symmetric free-boundary problem, of a dynamical MOTS saddle-node. The symmetry argument below captures the full principal eigenvalue, but the reported nonprincipal spectral gap remains restricted to the symmetric sector. The calculation does not identify a global horizon or establish a continuum bifurcation theorem.

## 2 Model and variational convention

### 2.1 Doubled cover and interval evolution

We use five-dimensional Einstein gravity with $\Lambda_5 = -6/\ell^2$ on the doubled $S^1/\mathbb{Z}_2$ cover. The numerical domain is one fundamental interval. With the Goldberger–Wise field $\Phi$ and collapse field $\chi$, the covering-space action is

$$S = \frac{1}{2\kappa_5^2}\int_{\widetilde{\mathcal{M}}} \mathrm{d}^5x\sqrt{-g}\,(R-2\Lambda_5) - \frac{1}{2}\int_{\widetilde{\mathcal{M}}} \mathrm{d}^5x\sqrt{-g}\,\left[(\nabla\Phi)^2 + m_\Phi^2\Phi^2 + (\nabla\chi)^2 + m_\chi^2\chi^2\right]$$
$$- \sum_i \int_{\Sigma_i} \mathrm{d}^4x\sqrt{-h_i}\,[\lambda_i + U_i(\Phi)], \qquad U_i(\Phi) = \frac{\gamma_i}{2}(\Phi - v_i)^2. \tag{1}$$

The branes are thin shells on the doubled cover; they are not ordinary outer boundaries of the action in Eq. (1). The evolution is performed on one side of each $\mathbb{Z}_2$ fixed surface. With outward normals from the fundamental interval, the one-sided junction conditions are

$$K^{(B)}_{\mu\nu} = \frac{\kappa_5^2}{2}\left(S_{\mu\nu} - \frac{1}{3}Sh_{\mu\nu}\right), \qquad n^a_{\text{out}}\nabla_a\Phi = -\frac{1}{2}U_i'(\Phi), \tag{2}$$

where $K^{(B)}_{\mu\nu}$ is the brane extrinsic curvature. An equivalent one-sided interval variational principle contains the appropriate Gibbons–Hawking–York terms and doubles the one-sided bulk contribution relative to each brane action [24, 25]. The thin-shell normalization follows the Israel junction formalism [26]. The junction conditions in Eq. (2) reproduce the tuned Randall–Sundrum signs at both walls. This is the convention implemented in the initial-data and evolution boundary rows.

Table 1: Dimensionless model and primary numerical parameters.

| quantity | value |
|---|---|
| $\epsilon$, backreaction, $m_\Phi^2\ell^2$ | 0.1, 0.01, 0.41 |
| wall stiffness $\gamma\ell$ | 20 |
| $m_\chi^2\ell^2$ | 0 |
| $A$, $\sigma_r/\ell$, $\sigma_y$, $y_c/d$ | 7.90, 1, 0.2, 0.9 |
| repaired critical-sequence domain | $R_{\max}/\ell = 10$ |
| repaired critical grids G9/G10/G11 | $113 \times 211$, $129 \times 241$, $145 \times 271$ |
| repaired evolution step | $\Delta t/\ell = 3.125 \times 10^{-5}$ |
| independent G10 temporal check | $\Delta t/\ell = 1.5625 \times 10^{-5}$ |

### 2.2 Normalization and initial family

All reported quantities are nondimensionalized by the AdS radius. We set $\ell = \kappa_5 = 1$, take $d/\ell = 1$, and use

$$z \in [1, e], \qquad r/\ell \in [0, R_{\max}/\ell]. \tag{3}$$

For the load-bearing repaired G9/G10/G11 line, $R_{\max}/\ell = 10$. The superseded pre-repair G7/G8 line retained only as a historical record in the supplement used $R_{\max}/\ell = 8$. Thus the quoted time is $t/\ell$. The background parameters and primary discretizations are listed in Table 1.

Writing $y = \log z$ and $d = 1$, the collapse field is initialized as

$$\chi(z, r) = A\, P_N(y; y_c, \sigma_y, d) \exp\left(-\frac{r^2}{2\sigma_r^2}\right), \tag{4}$$

where $P_N$ is the normalized even periodic-image sum that enforces Neumann data at both compact walls. Five image pairs are retained. Therefore $A = 7.90$ in Eq. (4) is the peak coefficient of a completely specified pulse family, not an invariant critical parameter. Initial data are momentarily time symmetric,

$$K_{ij} = 0, \qquad \Pi_\Phi = 0, \qquad \Pi_\chi = 0, \tag{5}$$

and the coupled Hamiltonian/stabilizer system is solved before evolution.

## 3 Capped MOTSs and the free-boundary stability problem

Let $u^a$ be the future unit normal to a spatial slice $M$, and let $N^a$ be the outward unit normal to a candidate three-surface $\mathcal{S} \subset M$. For $\ell_+^a = u^a + N^a$, our outgoing expansion is

$$\theta_+ = D_i N^i + K_{ij} N^i N^j - K. \tag{6}$$

A MOTS satisfies $\theta_+ = 0$ in Eq. (6). The upper-brane-capped class consists of $SO(3)$-symmetric polar graphs

$$z = z_B - \rho(\vartheta)\cos\vartheta, \qquad r = \rho(\vartheta)\sin\vartheta, \qquad 0 \le \vartheta \le \frac{\pi}{2}, \tag{7}$$

with regular closure at the axis and orthogonal contact with the upper brane.

### 3.1 Physical Robin condition and its coordinate reduction

The free-boundary perturbation condition is derived rather than imposed as a coordinate endpoint rule. Let $\varrho$ be the outward unit normal to the brane within $M$, let $\nu$ be the outward conormal to $\partial\mathcal{S} \subset \mathcal{S}$, and let $N$ be the unit surface normal. At orthogonal contact,

$$\nu = \varrho, \qquad g(N, \varrho) = 0. \tag{8}$$

For a variation with physical normal component $fN$, linearizing the contact condition gives the standard free-boundary operator

$$Bf := \nabla_\nu f - \mathrm{II}^{\partial M}(N, N) f = 0 \qquad \text{on } \partial\mathcal{S}, \tag{9}$$

where $\mathrm{II}^{\partial M}(X, Y) = g(\nabla_X \varrho, Y)$ [18, 20]. The brane curvature term is therefore not optional.

It is nevertheless already contained in the graph variable used by the calculation. The upper-brane cap terminates at the upper wall $z = e$. We orient the wall normal $\varrho$ out of the retained interval, the surface conormal $\nu$ outward from $\mathcal{S}$, and the MOTS normal $N$ toward increasing areal radius. Wall parity gives $h_{zr} = 0$ there, and hence $N = h_{rr}^{-1/2}\partial_r$ and $\nu = \varrho = h_{zz}^{-1/2}\partial_z$ at the cap endpoint. A graph displacement $\delta\rho$ has physical normal amplitude

$$f = w\,\delta\rho, \qquad w = g(N, \partial_r) = \sqrt{h_{rr}}. \tag{10}$$

The same wall geometry gives

$$\mathrm{II}^{\partial M}(N, N) = \frac{1}{2}\nu(\log h_{rr}) = \nu(\log w). \tag{11}$$

Substituting Eq. (11) into Eq. (9) yields

$$Bf = w\,\nu(\delta\rho). \tag{12}$$

Consequently, the implemented endpoint condition $\partial_\vartheta \delta\rho = 0$, equivalently $\partial_\vartheta(f/w) = 0$, is the coordinate form of the physical Robin condition for the wall-orthogonal graphs in Eq. (7); it does not discard the brane-extrinsic-curvature term. The same zero derivative at $\vartheta = 0$ is the smooth-axis regularity condition.

The stability operator is

$$Lf = \delta_{fN}\theta_+, \qquad Bf = 0. \tag{13}$$

The problem in Eq. (13) is evaluated on the complete evolved slice and is generally non-self-adjoint [11]. With our sign convention, a nonnegative principal eigenvalue is outward-stable. The operator and its Robin domain commute with the $SO(3)$ action. The elliptic principal eigenfunction is positive and unique up to scale; applying any group element therefore reproduces the same normalized eigenfunction. It follows that the principal eigenfunction is $SO(3)$-invariant and the symmetric calculation captures the full principal eigenvalue and its simple kernel. This argument does not determine the nonprincipal nonsymmetric spectrum. The matrix is a centered physical-normal Fréchet linearization after seven-point local finite-difference elimination of the endpoint Robin conditions. The boundary treatment is checked by the manufactured Robin spectrum, endpoint defects below $10^{-10}$, and 49/65/81 angular-node sequences; no alternative stencil-width study is claimed.

# 4 Numerical strategy

The coupled Einstein–scalar equations are evolved in a regular $SO(3)$-reduced generalized-harmonic formulation [27]. Compact-wall rows own the Israel and scalar conditions, interior-axis rows own regularity, and the wall–axis and wall–outer corners use a fixed staged ownership policy. A second wall-owner policy is retained as a control. The surface analysis is performed only after finite-value, Lorentzian-signature, constraint, wall/axis, and endpoint checks pass.

The load-bearing repaired-line detector solves the local second-order cap equation as an adaptive two-point boundary-value problem with $\rho'(0) = \rho'(\pi/2) = 0$ from the fixed seed bank

$$1.15, 1.20, \ldots, 1.70. \tag{14}$$

Admission uses the local residual, endpoint slope, physical domain, and an independent expansion cross-evaluator. The $10^5$ condition-number threshold reported in the supplement belongs only to a spectral detector used on the superseded pre-repair line. It is not applied to the repaired direct search, the local BVP continuation, or the augmented pseudo-arclength corrector. The principal stability operator whose eigenvalue crosses zero is likewise not that spectral detector matrix. In particular, singularity of the stability operator at the fold does not imply that the unrelated historical detector gate is triggered. No continuation condition-number trace was archived, so no conditioning trend is asserted or used as evidence.

The load-bearing critical calculation uses an independently constructed, parity-repaired parent sequence on G9, G10, and G11. A direct search is empty at $t/\ell = 0.001$ and admits exactly two branches at $t/\ell = 0.0015$ on all three grids. From the outer branch at the latter time, the local cap equation is continued backward with time promoted to an unknown and an integral pseudo-arclength condition appended [28]. The spacetime between stored RK2 endpoints is supplied by the fixed second-order dense extension of the evolution method; every endpoint replays bitwise. Corrector failures trigger only a prespecified dyadic arclength backoff.

After the principal eigenvalue changes sign, the arclength interval is halved until the time bracket is below $\Delta t/64$. At the smaller-magnitude endpoint of the final bracket, the physical-normal stability matrix is evaluated at 49, 65, and 81 angular nodes and repeated at a doubled Fréchet step. Because the operator is non-self-adjoint, its discrete left and right modes require an explicit convention. On the interior uniform angular nodes, the right eigenvector is phased so that its largest component is positive real and scaled to unit $\ell_\infty$ norm; the Euclidean conjugate-transpose left eigenvector is then scaled so that $f_L^* f_R = 1$. No mass matrix or continuum quadrature enters this normalization. The reported coefficient magnitudes are therefore reproducible near-critical discrete convention values; their resolved nonvanishing, rather than invariance under rescaling, is the normal-form test. Any admissible change of eigenvector normalization rescales $a$ and $b$ by finite nonzero factors and therefore cannot change whether either coefficient vanishes. The two local coefficients are evaluated as

$$a = \langle f_L, \partial_t \theta_+ \rangle, \qquad b = \frac{1}{2} \left\langle f_L, \partial^2_{f_R} \theta_+ \right\rangle, \tag{15}$$

with centered step-halving controls; these are the standard fold nondegeneracy quantities [29]. Finally, both branches are independently resolved at five prespecified time offsets and $(\mathcal{A}_{\rm out} - \mathcal{A}_{\rm in})^2$ is fitted linearly in time.

The load-bearing initial statement is explicitly search-bounded. On the G9 and G10 constraint-solved initial slices, regular-axis shooting uses 2,049 uniform launch radii covering

$$0.10 \le \rho_0/\ell \le 1.67, \tag{16}$$

together with twelve cutoff, tolerance, and step variants. Opposite-wall caps, closed star-shaped surfaces, and interval-spanning surfaces are searched separately in their frozen bounded classes. The

Table 2: Direct three-grid pair geometry and stability at $t/\ell = 0.0015$. $\mathcal{A}$ is the one-sided proper cap area.

| grid | $\mathcal{A}_{\text{in}}$ | $\lambda_{\text{in}}$ | $\mathcal{A}_{\text{out}}$ | $\lambda_{\text{out}}$ |
|---|---|---|---|---|
| G9 | 40.67520320 | −0.25894888 | 40.72845794 | +0.29347891 |
| G10 | 40.67520099 | −0.25895908 | 40.72845823 | +0.29348687 |
| G11 | 40.67519977 | −0.25896247 | 40.72845818 | +0.29348984 |

separate $A = 7.94$ positive control belongs to the superseded pre-repair study and is not used to qualify this initial null result.

# 5 Results

## 5.1 Three-grid zero-to-two formation

The independently constructed repaired critical sequence contains no accepted upper-brane-capped branch at $t/\ell = 0.001$ and exactly two at $t/\ell = 0.0015$ on G9, G10, and G11:

$$0.001 < t_{\text{form}}/\ell \leq 0.0015. \tag{17}$$

Equation (17) is a sampled formation bracket, not the later continuation estimate. Each branch is recovered from six seeds. Table 2 shows the direct geometric and stability observables. The adjacent-grid relative differences are at most $1.02 \times 10^{-6}$ for either endpoint, $1.76 \times 10^{-7}$ for the proper geometry, and $3.94 \times 10^{-5}$ for the principal eigenvalue. The negative inner and positive outer eigenvalues therefore close directly across three spacetime grids, rather than being inferred from convergence of the underlying fields.

## 5.2 Continuation to the critical surface

Backward pseudo-arclength continuation follows the stable outer branch around the turn and onto the unstable inner branch. It encloses the principal zero in time intervals of width $1.84 \times 10^{-7}$ (G9), $1.72 \times 10^{-7}$ (G10), and $1.87 \times 10^{-7}$ (G11). To the precision supported by the temporal check below, each gives

$$\frac{t_*}{\ell} \simeq 1.17254 \times 10^{-3}. \tag{18}$$

The gridwise values summarized in Eq. (18) have full spread $9.23 \times 10^{-9}$ in time and $8.55 \times 10^{-5}$ in the selected near-critical one-sided area. Both sequences are nonmonotone, as are the coefficient sequences below, so they are reported as finite-resolution scatter rather than assigned a Richardson order or a continuum extrapolation. The representative near-critical one-sided cap area is $\mathcal{A} \simeq 40.7034$.

Figure 1 displays the resolved branch structure. Proper area and principal eigenvalue separate with opposite signs on the two sides, while the area separation follows the square-root law on every grid.

## 5.3 Independent half-timestep comparison

The spatial sequence shares a second-order RK2 step and dense extension, so its grid agreement alone cannot bound a common temporal bias. We therefore reconstructed the G10 parent from $t/\ell = 0.001$ to 0.0015 with $\Delta t/(2\ell) = 1.5625 \times 10^{-5}$, independently re-solved both endpoint anchors, replayed the full half-step trajectory, and repeated the continuation, stability, coefficient, and area-scaling

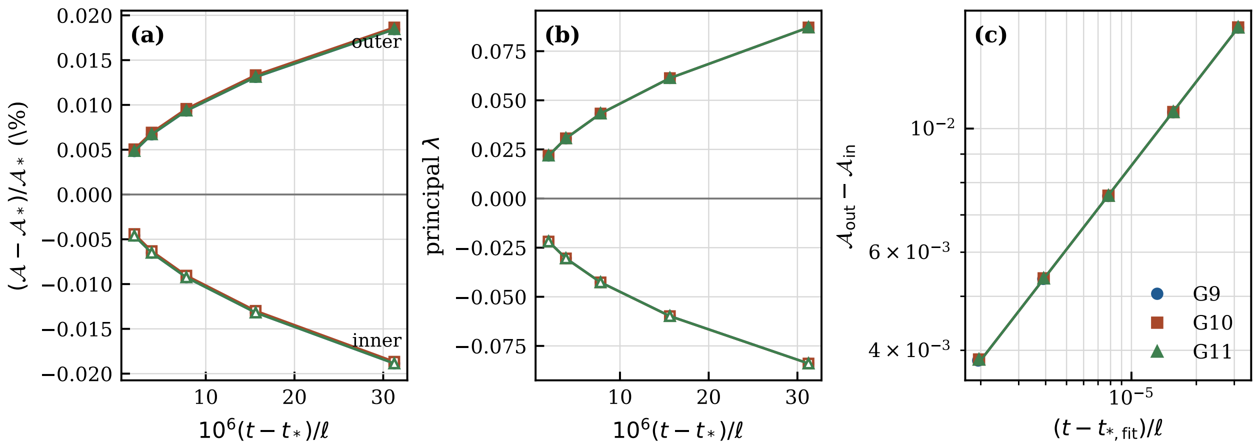


Figure 1: Three-grid resolution of the free-boundary MOTS saddle-node. **(a)** One-sided proper cap area relative to the gridwise near-critical reference area $\mathcal{A}_*$ at the selected bracket endpoint, not an exact critical value; filled symbols denote the outer branch and open symbols the inner branch. **(b)** The corresponding principal stability eigenvalues, which approach zero with opposite signs. **(c)** Log–log area separation and the gridwise fits $\Delta\mathcal{A} = [s(t-t_{*,\rm fit})]^{1/2}$.

Table 3: G10 half-timestep comparison. The $t_*$ entries are bracket midpoints; their difference is not an uncertainty estimate. Differences for $\mathcal{A}_{\rm nc}$, $a$, and $b$ are relative; $\mathcal{A}_{\rm nc}$ is the selected near-critical reference area.

| quantity | $\Delta t/\ell$ | $\Delta t/(2\ell)$ | difference |
|---|---:|---:|---:|
| $t_*/\ell$ | 0.0011725452 | 0.0011725443 | $9.17\times10^{-10}$ |
| $\mathcal{A}_{\rm nc}$ | 40.703366 | 40.703400 | $8.45\times10^{-7}$ |
| $a$ | $-20.76565$ | $-20.76535$ | $1.42\times10^{-5}$ |
| $b$ | 2.80410 | 2.80435 | $9.23\times10^{-5}$ |
| free exponent $p$ | 0.49597 | 0.49806 | — |
| $R^2$ for squared-area fit | 0.9999804 | 0.9999951 | — |
| zero-bracket width | $1.72\times10^{-7}$ | $1.79\times10^{-7}$ | — |

calculations. Table 3 shows the direct comparison. All criteria were fixed before the half-step result was examined.

The full- and half-step critical-time brackets overlap. Their midpoint shift is $9.17\times10^{-10}$, while the brackets themselves remain approximately $1.8\times10^{-7}$ wide; the midpoint shift is therefore not interpreted as the uncertainty in $t_*$. The half-step run retains the isolated near-critical principal mode, coefficient signs, and square-root scaling. The measured changes satisfy the prospective ceilings: $\Delta t/64$ for the bracket-midpoint shift, 1% for the near-critical area, and 20% for each coefficient. The prescribed decision rule therefore strongly constrains the shared temporal-discretization concern without triggering a quarter-step calculation. Although this is not a separate convergence study of constraint violation, the complete half-step evolution independently passes the same native constraint and boundary gates while reproducing the critical observables. It therefore supplies an indirect temporal check against systematic evolution error rather than only a continuation-level comparison.

Table 4: Near-critical discrete normal-form observables. Coefficients are evaluated at the smaller-magnitude endpoint of the final zero bracket; the final entries of the two-step evaluations are shown.

| quantity | G9 | G10 | G11 |
|---|---|---|---|
| next reduced-sector eigenvalue | 5.56598 | 5.56531 | 5.56595 |
| near-critical transversality $a$ | $-20.76513$ | $-20.76565$ | $-20.76493$ |
| near-critical quadratic coefficient $b$ | 2.80717 | 2.80410 | 2.81060 |
| square-root exponent $p$ | 0.4959 | 0.4960 | 0.4960 |
| $R^2$ for $(\Delta\mathcal{A})^2$ fit | 0.9999801 | 0.9999804 | 0.9999803 |

### 5.4 Near-critical mode, nondegeneracy, and invariant scaling

Table 4 gives the near-critical diagnostics. The representative profile is the smaller-magnitude endpoint of the final opposite-sign bracket, so its listed eigenvalue need not vanish identically. At those representatives, $|\lambda|$ lies between $7.4 \times 10^{-4}$ and $1.7 \times 10^{-3}$. The next eigenvalue in the $SO(3)$-reduced discretization remains near 5.566, leaving a large reduced-sector spectral gap. The principal right mode is real and sign-definite. The unit left–right overlap is imposed by the normalization above and is not counted as independent evidence.

The centered step-halving evaluations of Eq. (15) retain $a < 0$ and $b > 0$ on every grid. Adjacent relative differences are at most $3.46 \times 10^{-5}$ for $a$ and $2.32 \times 10^{-3}$ for $b$, compared with the prospective 0.20 ceilings. Together, the isolated near-critical principal mode and its bracketed zero crossing are consistent with a simple critical kernel; the temporal crossing is transverse and the quadratic term is resolved and nonzero under the stated discrete convention. Independently, five paired fixed-time solves per grid give

$$\mathcal{A}_{\rm out} - \mathcal{A}_{\rm in} \propto (t - t_*)^p, \qquad p \simeq 0.496 \tag{19}$$

Equation (19) holds on the three-grid sequence, and $p \simeq 0.498$ in the half-step G10 run. The quoted $R^2 = 0.9999801$–$0.9999804$ belongs to the linear fit of $(\Delta\mathcal{A})^2$ against time, not to the separate free-exponent fit. Fixing $t_*$ to the continuation midpoint gives $p = 0.4951$–$0.4964$; omitting one of the five offsets gives $p = 0.4919$–$0.4996$. A fixed $p = 1/2$ fit has maximum relative residual 0.9–1.2%. These sensitivity checks support a near-one-half law but do not justify treating the last exponent digits as an uncertainty estimate. The shift from $p \simeq 0.496$ at the full step to $p \simeq 0.498$ at the half step is toward one-half, consistent with a residual temporal-discretization bias rather than a fit-selection artifact; two time steps are not used to infer an error order. The continuation, operator, adjoint, and invariant-geometry tests therefore identify the local saddle-node structure through mutually distinct observables.

## 6 Interpretation

The calculation resolves the local data that distinguish a generic saddle-node from a merely suggestive zero-to-two history. The branches are connected by pseudo-arclength continuation; an isolated near-critical principal mode accompanies the bracketed zero crossing; both the adjoint temporal transversality and quadratic coefficient are nonzero; and an invariant geometric separation has the expected one-half exponent. None of these ingredients is inferred from the branch count alone. Their agreement across three spacetime grids and the independent half-timestep G10 comparison support a numerical free-boundary MOTS saddle-node at finite resolution in the imposed $SO(3)$-symmetric problem.

The physical significance of the result lies in the surface class and arena. The MOTSs terminate orthogonally on a brane inside an explicitly stabilized compact interval. They are not imposed as static horizons and are not continued from a pre-existing connected black string. Their appearance is consistent with the local paired-MOTS mechanism under the physical Robin boundary condition in the tested nonlinear evolution. The saddle-node mechanism itself is established theory; the contribution here is resolving its critical structure in a dynamical brane-terminating problem.

# 7 Limitations and boundary of inference

The initial null result is bounded by the stated launch interval, seed family, and searched surface classes. MOTSs depend on slicing; no repaired-line gauge family is surveyed here. The critical time is a finite-resolution continuation estimate in the chosen foliation. The half-timestep comparison constrains the common temporal bias, but the nonmonotone three-grid scatter is not promoted to a continuum theorem, Richardson order, or formal uncertainty interval. Equivariance and uniqueness place the principal mode in the $SO(3)$-invariant sector, but the nonprincipal spectral-gap statement and the discrete normal-form evaluation remain tied to the symmetric branch and stated discretization. They are numerical results, not analytic Fredholm bounds on the uncomputed nonsymmetric nonprincipal spectrum. Separately solved larger-domain parents do not define the same universal formation datum. Finally, no event horizon, common horizon, global topology change, or nonlinear stability theorem follows from the evidence.

These are not withheld conclusions from the same calculation. Each requires a different observable or a different global construction.

# 8 Conclusion

Localized collapse in a Goldberger–Wise-stabilized compact braneworld produces a brane-terminating free-boundary MOTS saddle-node. An independently constructed parent sequence evolves from no accepted capped surface to two opposite-stability branches on G9, G10, and G11. Pseudo-arclength continuation connects them through a grid-consistent critical surface. The isolated near-critical principal mode and bracketed zero crossing are consistent with a simple critical kernel; the adjoint transversality and quadratic coefficients are nonzero under the stated discrete normalization, and the invariant proper-area separation follows a near-one-half law. An independent G10 half-timestep evolution strongly constrains the shared temporal-discretization concern. The repaired-line residual, operator-resolution, physical-boundary, and deterministic-replay controls support the formation and critical-structure result. The calculation therefore advances from observing post-formation branch anatomy to numerically resolving the local critical structure of a dynamical free-boundary MOTS problem. A separate future study may use these leaves to investigate a marginal tube or other global horizon observables; none is claimed here.

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## Data and code availability

Machine-readable results, frozen protocols, source manifests, tests, and the scripts supporting the figures are archived at Zenodo under doi:10.5281/zenodo.22085055.

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